

**KIT- KALAIGNARKARUNANIDHI INSTITUTE OF TECHNOLOGY**  
**(An Autonomous Institution, Affiliated to Anna University Chennai)**

Coimbatore-641 402

**Question Bank**

|                     |   |                            |               |   |        |
|---------------------|---|----------------------------|---------------|---|--------|
| Department          | : | CSE(AI&ML)                 | Year/Semester | : | 3/EVEN |
| Subject Code & Name | : | B23AMI602 & Machine Vision | Regulations   | : | 2023   |

| UNIT - I          |                                                                                                                         |       |     |    |
|-------------------|-------------------------------------------------------------------------------------------------------------------------|-------|-----|----|
| Part-A (2 Marks)  |                                                                                                                         |       |     |    |
| Q. No.            | Questions                                                                                                               | CO    | K   |    |
| 1                 | Explain the term Machine vision                                                                                         | CO1   | K2  |    |
| 2                 | Summarize the differences between human vision and machine vision.                                                      | CO1   | K2  |    |
| 3                 | Explain the concept of light–matter interaction.                                                                        | CO1   | K2  |    |
| 4                 | State the thin lens equation..                                                                                          | CO1   | K2  |    |
| Part-B (13 Marks) |                                                                                                                         |       |     |    |
| Q. No.            | Questions                                                                                                               | Marks | CO  | K  |
| 1                 | Explain the differences between human vision, machine vision, and computer vision with suitable examples.               | 13    | CO1 | K2 |
| 2                 | Explain the block diagram of a machine vision system and describe the function of each component.                       | 13    | CO1 | K2 |
| 3                 | Explain the benefits of machine vision systems in industrial applications.                                              | 13    | CO1 | K2 |
| 4                 | Explain the physics of light, light–matter interactions, refraction at a spherical surface, and the thin lens equation. | 13    | CO1 | K2 |

| Unit II -         |                                                                                                     |       |     |    |
|-------------------|-----------------------------------------------------------------------------------------------------|-------|-----|----|
| Part-A (2 Marks)  |                                                                                                     |       |     |    |
| Q. No.            | Questions                                                                                           | CO    | K   |    |
| 1                 | Apply scene constraints to improve image acquisition in a machine vision system.                    | CO2   | K3  |    |
| 2                 | Apply lighting parameters to enhance image quality.                                                 | CO2   | K3  |    |
| 3                 | Apply the concept of CCD/CMOS sensors in image capturing.                                           | CO2   | K3  |    |
| 4                 | Apply optical filters to reduce noise in images..                                                   | CO2   | K3  |    |
| Part-B (13 Marks) |                                                                                                     |       |     |    |
| Q. No.            | Questions                                                                                           | Marks | CO  | K  |
| 1                 | Apply scene constraints and lighting parameters to design an effective image acquisition setup.     | 13    | CO2 | K3 |
| 2                 | Apply different lighting techniques and justify their selection for a given industrial application. | 13    | CO2 | K3 |

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|   |                                                                                                            |    |     |    |
|---|------------------------------------------------------------------------------------------------------------|----|-----|----|
| 3 | Apply lens specifications and optical filters to achieve proper image formation in machine vision systems. | 13 | CO2 | K3 |
| 4 | Apply image sensor selection, camera types, and interface architectures for an industrial vision system.   | 13 | CO2 | K3 |

**Unit III -**

**Part-A (2 Marks)**

| Q. No. | Questions                                                            | CO  | K  |
|--------|----------------------------------------------------------------------|-----|----|
| 1      | Apply digital image fundamentals in image processing.                | CO3 | K3 |
| 2      | Demonstrate thresholding for segmenting an image.n.                  | CO3 | K3 |
| 3      | Use smoothing techniques to remove noise from an image.              | CO3 | K3 |
| 4      | Show how edge detection is applied in identifying object boundaries. | CO3 | K3 |

**Part-B (13 Marks)**

| Q. No. | Questions                                                                                                | Marks | CO  | K  |
|--------|----------------------------------------------------------------------------------------------------------|-------|-----|----|
| 1      | Implement image acquisition modes and machine vision software concepts in a practical system.            | 13    | CO3 | K3 |
| 2      | Demonstrate the use of spatial and frequency domain techniques for image processing..                    | 13    | CO3 | K3 |
| 3      | Solve image enhancement problems using point operations, thresholding, and grayscale transformations..   | 13    | CO3 | K3 |
| 4      | Use smoothing, sharpening, edge detection, and morphological operations in an image processing workflow. | 13    | CO3 | K3 |

**Unit IV –**

**Part-A (2 Marks)**

| Q. No. | Questions                                                   | CO  | K  |
|--------|-------------------------------------------------------------|-----|----|
| 1      | Differentiate region features from shape features           | CO4 | K4 |
| 2      | Examine the characteristics of size and shape features      | CO4 | K4 |
| 3      | Analyze texture variations in images                        | CO4 | K4 |
| 4      | Compare template matching with other recognition techniques | CO4 | K4 |

**Part-B (13 Marks)**

| Q. No. | Questions                                                                                       | Marks | CO  | K  |
|--------|-------------------------------------------------------------------------------------------------|-------|-----|----|
| 1      | Analyze various feature extraction techniques based on region, shape, and size characteristics. | 13    | CO4 | K4 |

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|   |                                                                                           |    |     |    |
|---|-------------------------------------------------------------------------------------------|----|-----|----|
| 2 | Examine texture analysis methods and their effectiveness in machine vision.               | 13 | CO4 | K4 |
| 3 | Differentiate template matching and classification techniques used in object recognition. | 13 | CO4 | K4 |
| 4 | Analyze 3D machine vision techniques and categorize decision-making strategies.           | 13 | CO4 | K4 |

**Unit V –**

**Part-A (2 Marks)**

| Q. No. | Questions                                                       | CO  | K  |
|--------|-----------------------------------------------------------------|-----|----|
| 1      | Examine the role of machine vision in manufacturing industries. | CO5 | K4 |
| 2      | Differentiate OCR and OCV systems.                              | CO5 | K4 |
| 3      | Analyze the working of vision-guided robots                     | CO5 | K4 |
| 4      | Examine biometric systems using machine vision.                 | CO5 | K4 |

**Part-B (13 Marks)**

| Q. No. | Questions                                                                                                | Marks | CO  | K  |
|--------|----------------------------------------------------------------------------------------------------------|-------|-----|----|
| 1      | Analyze machine vision applications across manufacturing, electronics, printing, and textile industries. | 13    | CO5 | K4 |
| 2      | Examine applications in non-visible spectrum, metrology, and gauging systems.                            | 13    | CO5 | K4 |
| 3      | Differentiate OCR, OCV, and vision-guided robotics based on performance and usage.                       | 13    | CO5 | K4 |
| 4      | Analyze machine vision applications in agriculture, biomedical, surveillance, and biometrics.            | 13    | CO5 | K4 |